

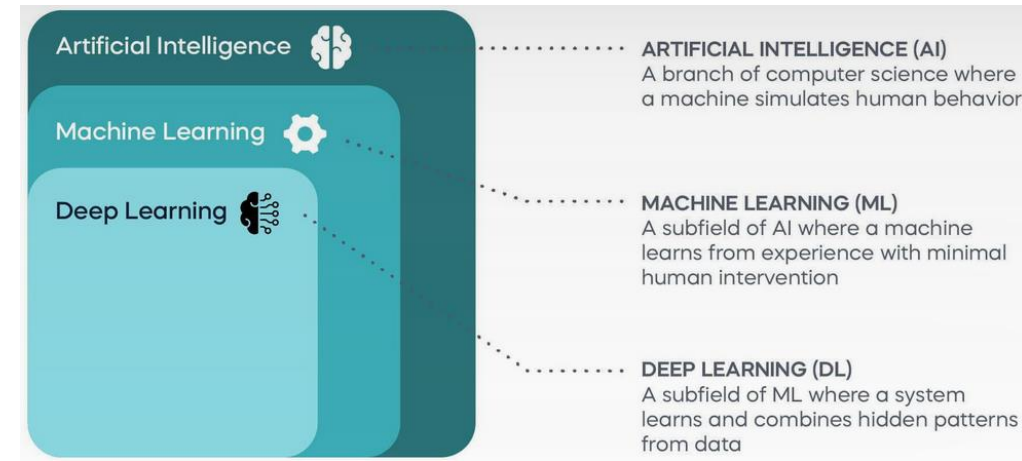
Data Analysis and Machine Learning

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What is Machine Learning?

- Machine learning (ML) is a subfield of statistics (termed statistical learning) and artificial intelligence, also related to behavioral economics.
- Machine learning (ML) uses algorithms trained on data sets to *create or adjust models* capable of performing tasks that would otherwise only be possible for humans.



Why Machine Learning?

- Prediction vs explanation (Pre vs. Post)
- Model selection
- Bias-variance trade-off
- Bayesian: To blend data analysis with background professional knowledge
- Human heuristic vs. AI



Sebastian Raschka 

@rasbt



Maybe a hot take, but what about the following advice to the next gen:

Don't get an AI degree; the curriculum will be outdated before you graduate. Instead, study math, stats, or physics as your foundation, and stay current with AI through code-focused books, blogs, and papers.

Why Coding?

- Exact vs. approximate solutions
- Verification of theoretical models
- Making predictions
- To have practical works
- To have an advantage on your resume

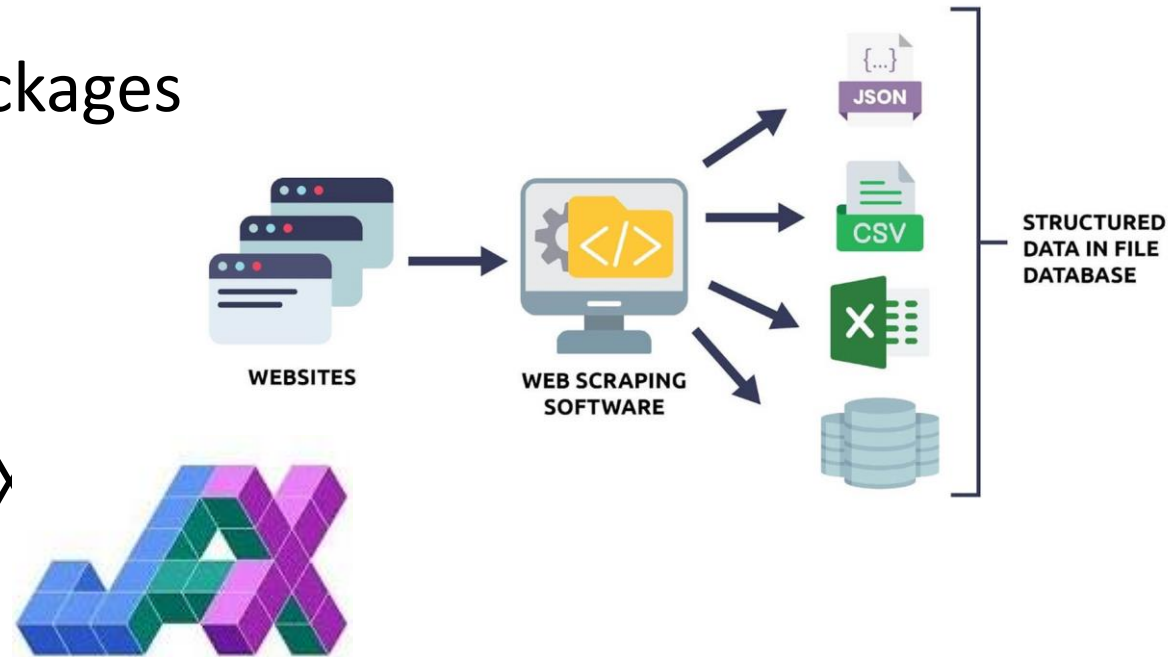
What is Python?

- The implementation of programming language Python was started in December 1989 by Guido van Rossum at CWI in the Netherlands.
- It was developed at CWI as a successor to the ABC language.
- Its name was inspired by the BBC comedy series Monty Python's Flying Circus.

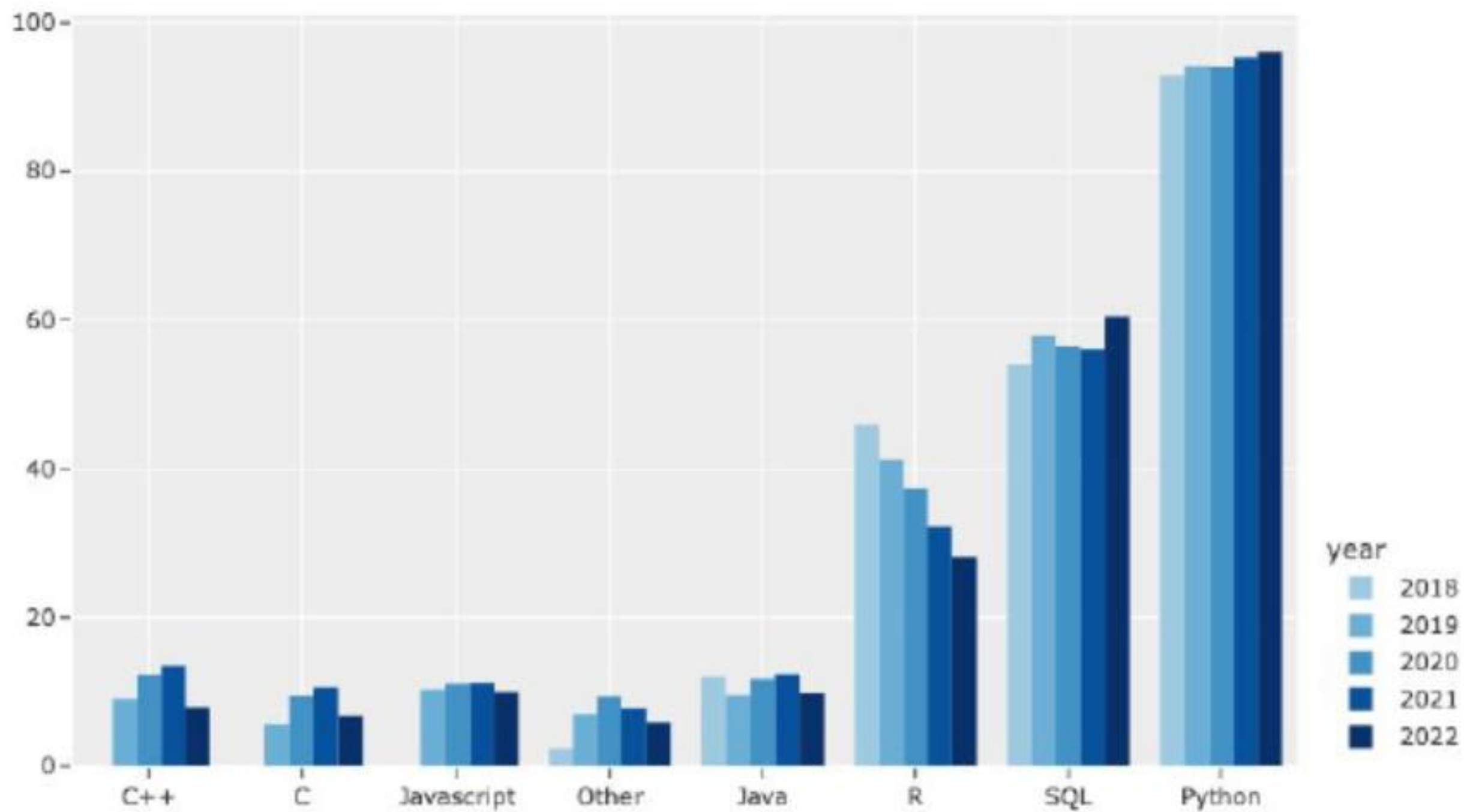


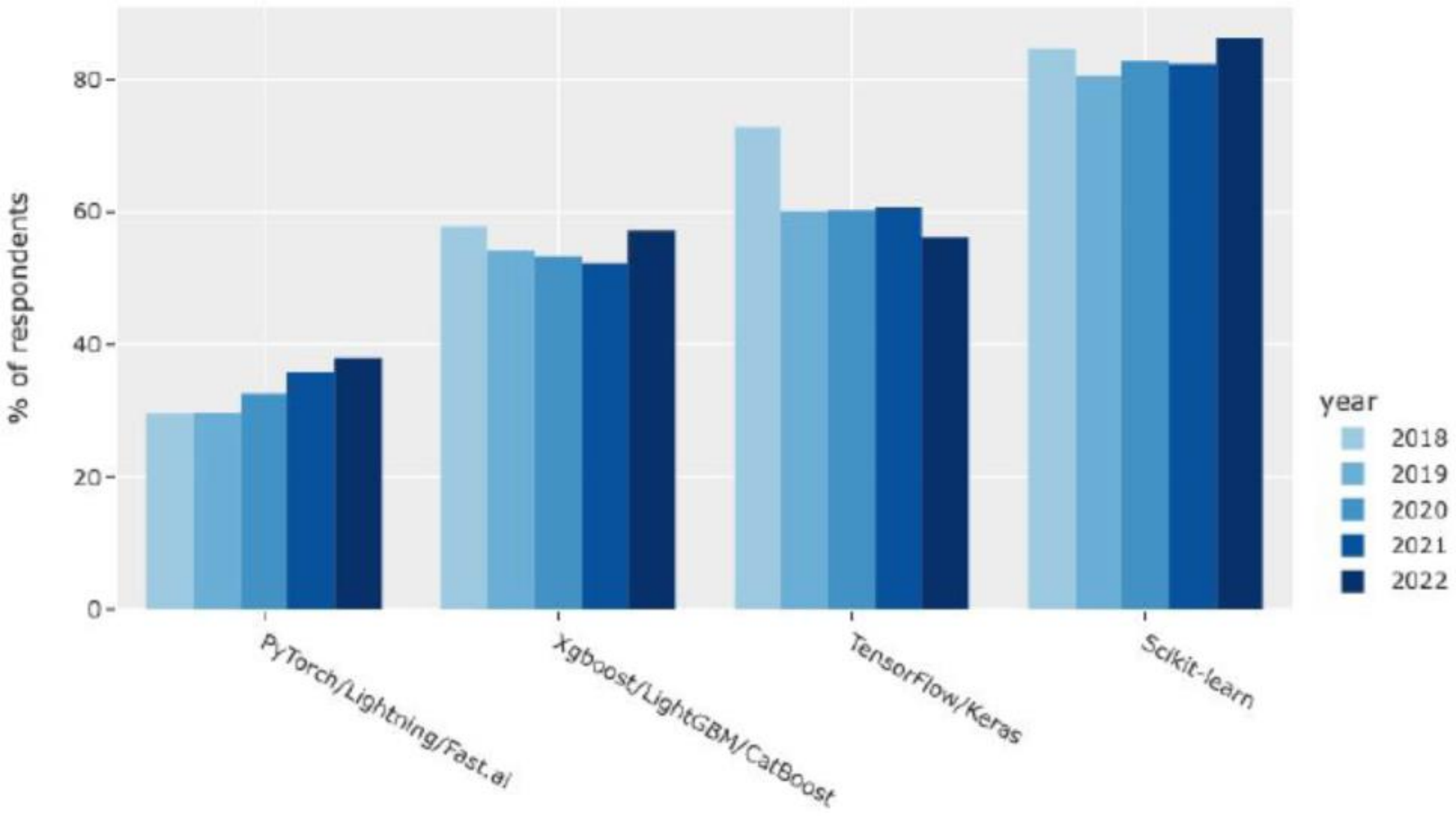
Why Python?

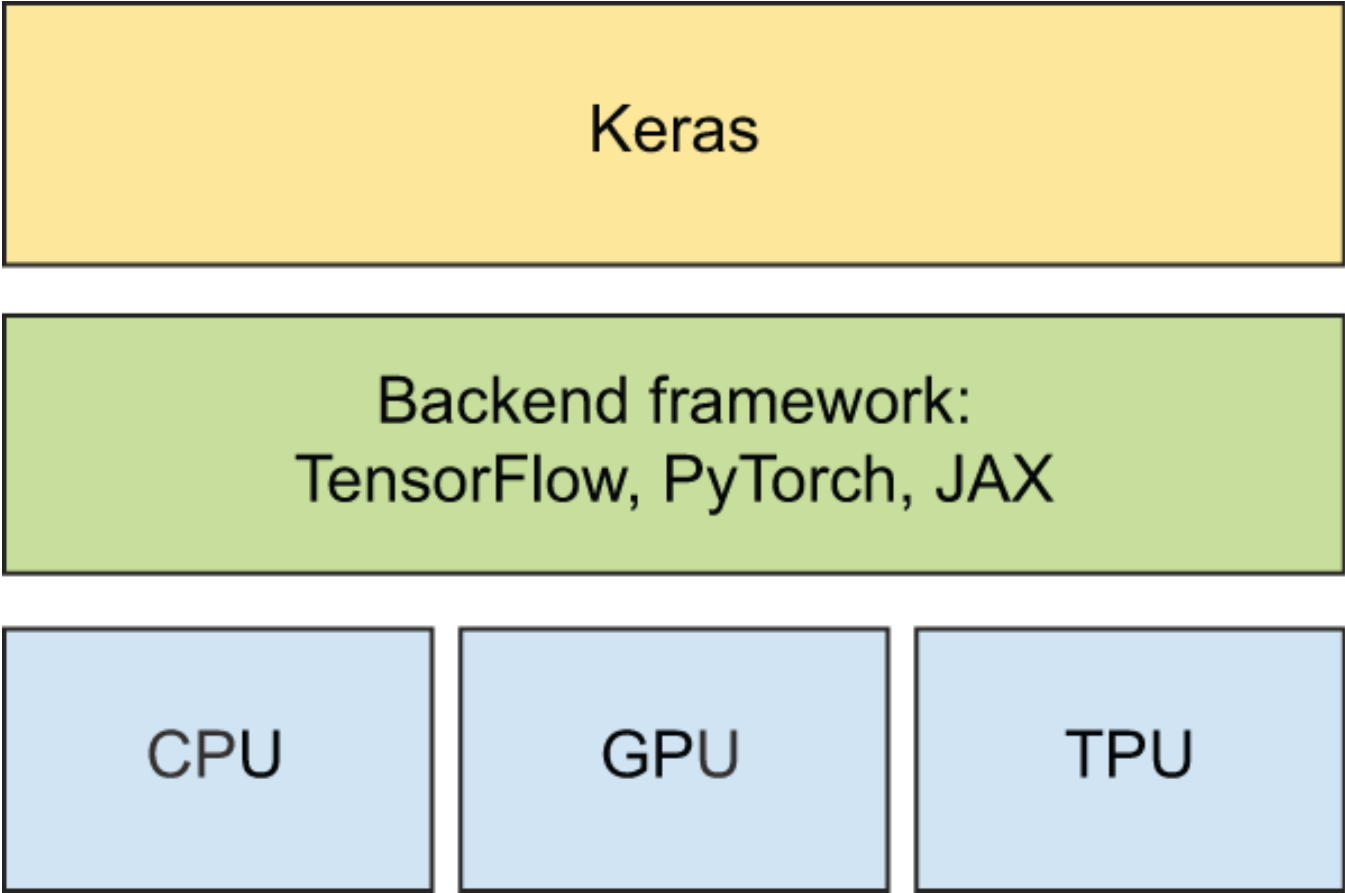
- Free
- Friendly for beginners
- Available resources and packages
- Powerful in web scraping
- Easy to extend (CUDA vs JA)



% of respondents







The diagram illustrates the layers of a deep learning stack. It consists of three main horizontal sections. The top section is a single yellow box labeled 'Keras'. The middle section is a single green box labeled 'Backend framework: TensorFlow, PyTorch, JAX'. The bottom section consists of three separate light blue boxes labeled 'CPU', 'GPU', and 'TPU' arranged horizontally. To the right of each section is a descriptive text block.

Keras

Deep learning development:
layers, models, optimizers, losses,
metrics, training loops ...

Backend framework:
TensorFlow, PyTorch, JAX

Tensor manipulation infrastructure:
tensors, variables, automatic
differentiation, distribution ...

CPU

GPU

TPU

Hardware: execution

黃仁勳點名3種人會先被AI淘汰

第一類，是沒有主見、只會照表操課的人。

這類工作者高度依賴既有流程，缺乏重新定義問題與目標的能力，一旦AI能更快、更準確地執行標準化任務，他們便失去存在價值。

第二類，是不懂提問、無法與AI深度互動的人。

黃仁勳直言，未來的核心能力不在於「下指令」，而在於「問對問題」。他甚至坦白表示：

「我90%的工作，其實都在提問。」無法提出高品質問題的人，即便擁有AI，也難以放大自身價值。

第三類，則是懶得學習，只把AI當成答案機的人。

黃仁勳指出，AI是一種「技術均衡器」，能讓非技術背景者成為創作者，但前提是——你必須知道要讓AI做什麼。若只依賴 AI給答案，卻不願學習、思考與判斷，最終仍會被流程取代。

Anaconda

- Sign up and sign in an account:
<https://www.anaconda.com/download>
- Download its distribution:
https://www.anaconda.com/download/success?user_state=logged_in&utm_campaign=direct_download&utm_medium=homepage_link&utm_source=platform
- This course uses the version: Anaconda3-2025.12-1-Windows-x86_64

Assessing Probability

- There are three viewpoints about how to assess the probability of an uncertain event:

- 1. Classical probability**

2. Relative frequency probability

3. Subjective probability

Classical Probability

- It is important to count the number of *all* possible outcomes (sample space).
- Then we count the number of outcomes which we are interested (event).
- Probability = $\#(\text{event}) / \#(\text{sample space})$

17th Century

- Pascal and Fermat's correspondence (1654 July)
- De Mere: "At least on 6 in 4 rolls" vs "at least on double-six in 24 throws of 2 dice" (crucial)

Truth is the same of Toulouse and at Paris. --by Pascal

Why Incident in Python

- Check an example:

```
i=1;  
for x in range(5):  
    print(x)  
i=i+1
```

What will `i` be in the end?

Assessing Probability

2. Relative-Frequency Probability (客觀的機率理論)

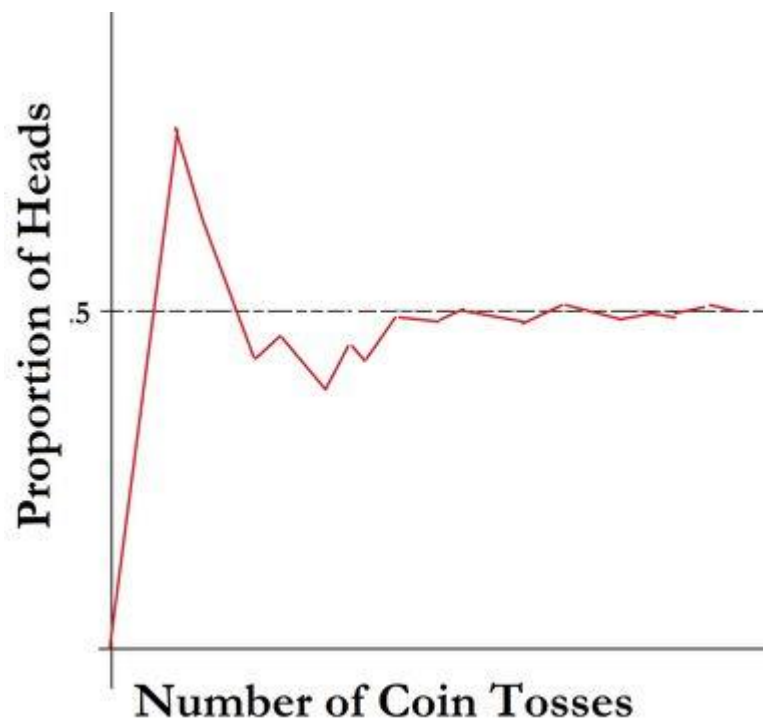
長期重複試行的隨機實驗中，事件的相對次數會漸趨穩定。而此一次數的比率即為該事件的機率。

$$P(E) = \lim_{n \rightarrow \infty} \frac{n(E)}{n}$$

即資料越多越接近真實值

Relative Frequency Probability

- 若某事件有既定的機率，而我們不斷的進行相同的實驗，則該事件發生的次數比率會越來越接近這個既定的機率。



Assessing Probability

3. Subjective probability

An individual opinion or belief about the probability of occurrence

EX. 對某市長候選人當選的可能性, 明天某股票股價會上漲的機率等。

Probability Axioms

1. If A is any event in the sample space S , then for each $E_i \in S$,

$$0 \leq P(E_i) \leq 1$$

2. Let E_i be the exclusive events in S . Then

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \dots + P(E_n)$$

3. $P(S) = 1$

Probability Rules

- The **Complement rule** (餘集合定理):

$$P(\bar{A}) = 1 - P(A) \quad \text{i.e., } P(A) + P(\bar{A}) = 1$$

Ex. De Mere's problem (1717)

- The **Addition rule** (加法定理):
 - The probability of the union of two events is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Monty Hall Problem Revisited

- Ex. Monty Hall problem (3-door problem)
Let's Make a Deal



Monty Hall Problem Revisited

- Ex. Monty Hall problem (3-door problem)

